ANALYSIS II — SHEET 5 (for Week 6 tutorials)
Uniform Convergence. Differentiability.

1. The functions $f, g, h\colon \mathbb{R} \to \mathbb{R}$ are defined by

$$f(x) = \begin{cases} x \sin \frac{1}{x} & x \neq 0, \\ 0 & x = 0, \end{cases} \qquad g(x) = \begin{cases} x^2 \sin \frac{1}{x} & x \neq 0, \\ 0 & x = 0, \end{cases} \qquad h(x) = \begin{cases} x^3 \sin \frac{1}{x} & x \neq 0, \\ 0 & x = 0. \end{cases}$$

(i) Show that f is continuous at 0, but not differentiable there.
(ii) Show that g is differentiable at 0, but that g' is not continuous there.
(iii) Show that h is differentiable at 0, that h' is continuous there, but h' is not differentiable there.

2. Consider the sequences (f_n) of functions on the interval $[0, 1]$ whose nth term $f_n(x)$ is defined as follows.

$$\text{(i)} \quad \left(\frac{x}{2}\right)^n, \quad \text{(ii)} \quad \sin(nx), \quad \text{(iii)} \quad \sqrt{n}\,(1-x)\,x^n, \quad \text{(iv)} \quad \begin{cases} \min\{x^{-1}, n\} & x \neq 0; \\ 0 & x = 0. \end{cases}$$

For each sequence, (i) determine the pointwise limit or show that it fails to exist, and (ii) if the convergence is pointwise, determine whether it is uniform also.

3. Consider the following series:

$$\sum_{k=0}^{\infty} \frac{\sin^2(kx)}{1 + k^2 x^2}.$$

(i) Prove that the series converges for each $x \in \mathbb{R}$ and denote its sum as $f(x)$.
(ii) Given $\delta > 0$, prove that the series converges uniformly on $(-\infty, -\delta] \cup [\delta, \infty)$. Deduce that f is continuous on $\mathbb{R}\backslash\{0\}$.
(iii) Show that f is not continuous at $x = 0$.

4. (i) Let $S \subseteq \mathbb{R}$, let $f_n\colon S \to \mathbb{R}$ be a sequence of continuous functions converging uniformly to $f\colon S \to \mathbb{R}$ and let (x_n) be a sequence in S converging to $p \in S$. Show that $f_n(x_n)$ converges to $f(p)$.

(ii) By quoting the power series of $-\log(1-x)$ centred at $x = 0$, or otherwise, carefully show that

$$\lim_{n\to\infty} \left[\sum_{k=1}^{n} \frac{1}{k} \left(\frac{1}{2} + \frac{1}{n} \right)^k \right] = \log 2.$$

5. (i) Evaluate

$$L = \lim_{x \to 1} \frac{\log x}{x - 1}$$

as a derivative.
(ii) Sketch the function $f\colon (0, \infty) \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{\log x}{x-1} & x \neq 1, \\ L & x = 1. \end{cases}.$$

(iii) Is f differentiable at $x = 1$?

6. Find all differentiable solutions $f\colon \mathbb{R} \to \mathbb{R}$ of the identity

$$\sin f(x) = \sin x \qquad \forall x \in \mathbb{R}.$$

Find a continuous, non-differentiable solution of the identity.

Extension Exercises (Optional)

7. The functions $f_n\colon (-1,1) \to \mathbb{R}$ are defined by $f_n(x) = |x|^{1+1/n}$. Show that:
(i) the functions f_n are differentiable.
(ii) (f_n) converges uniformly.
(iii) $\lim f_n$ is not differentiable.

8. Let $f\colon \mathbb{R} \to \mathbb{R}$ be differentiable at p. Say that $a_n < p < b_n$ with $a_n \to p$ and $b_n \to p$. Show that

$$f'(p) = \lim_{n\to\infty} \frac{f(b_n) - f(a_n)}{b_n - a_n}.$$

9. $\varphi\colon [0,1] \to \mathbb{R}$ is said to be a *step function* if there are finitely many intervals $I_k \subseteq [0,1]$ such that φ is constant on each I_k and zero otherwise.

(i) Let $f\colon [0,1] \to \mathbb{R}$ be a continuous function. Show that there is a sequence of step functions (φ_n) which converge uniformly to f.

(ii) The *Dirichlet function* $D\colon [0,1] \to \mathbb{R}$ is defined by

$$D(x) = \begin{cases} 1 & x \in \mathbb{Q}, \\ 0 & x \notin \mathbb{Q}. \end{cases}$$

Show that D is not the uniform limit of step functions.

(iii) Let $g\colon [0,1] \to \mathbb{R}$ be a function which is the uniform limit of step functions. Show that $g(c^+)$ and $g(c^-)$ exist for each $c \in (0,1)$.

(iv) Let $\varphi_n\colon [0,1] \to \mathbb{R}$ be a sequence of step functions which converge uniformly to the function $f(x) = x$. Let d_n denote the number of discontinuities of φ_n. Show that $d_n \to \infty$.

10. The function $g\colon \mathbb{R} \to \mathbb{R}$ is defined by $g(x) = |x|$ on $[-1,1]$ and then extended with period 2. We define the Weierstrass-like function $W\colon \mathbb{R} \to \mathbb{R}$ by

$$W(x) = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n g(4^n x).$$

(i) Sketch the zeroth, first and second partial sums.

(ii) Show that W is continuous.

(iii) Show, for any real x and positive integer m, that it is possible to choose $\delta_m = \pm 2^{-2m-1}$ so that the open interval with end-points $4^m x$ and $4^m(x + \delta_m)$ contains no integer.

(iv) By considering

$$\frac{W(x+\delta_m) - W(x)}{\delta_m},$$

or otherwise, show that W is not differentiable at x.

11. (i) Show that

$$\frac{1}{n}x\left(1-x\right)=\sum_{k=0}^{n}\left(x-\frac{k}{n}\right)^{2}\binom{n}{k}x^{k}\left(1-x\right)^{n-k}.$$

(ii) Given continuous $f\colon\ [0,1]\to\mathbb{R}$, the nth *Bernstein polynomial* is defined to be

$$B_{n}\left(x\right)=\sum_{k=0}^{n}f\left(\frac{k}{n}\right)\binom{n}{k}x^{k}\left(1-x\right)^{n-k}.$$

Note that

$$\left|f\left(x\right)-B_{n}\left(x\right)\right|\leqslant\sum_{k=0}^{n}\left|f\left(x\right)-f\left(\frac{k}{n}\right)\right|\binom{n}{k}x^{k}\left(1-x\right)^{n-k}.$$

In the above sum, the k can be separated into those where $|x-k/n|$ is suitably small, so that $|f\left(x\right)-f\left(k/n\right)|$ is small by continuity, and those k such that $|x-k/n|$ is larger so that $\binom{n}{k}x^{k}\left(1-x\right)^{n-k}$ is small.

Consider first, the above summands where $|x-k/n|>n^{-1/4}$. Show that their total contribution is less than $M/(2\sqrt{n})$ where M is a bound for $|f|$ on $[0,1]$. You will need (i) for this.

Find a bound for the remaining summands' contributions using uniform continuity.

Hence prove *Bernstein's Approximation Theorem*: given $\varepsilon>0$ there exists n such that

$$\left|f\left(x\right)-B_{n}\left(x\right)\right|<\varepsilon\qquad\text{for all }0\leqslant x\leqslant1.$$

(ii) Deduce *Weierstrass' Approximation Theorem* which states that, given a continuous function $g\colon\ [a,b]\to\mathbb{R}$, there is a sequence of polynomials which converge uniformly to g.